

The 12 Primitives

Glyph	Name	n	Subtypes	AA
Đ	Dimensionality	4	↳↵↻↻	Gly
Ɔ	Topology	5	↵↻↻↻↻	Pro
Ř	Recognition	4	↻↻↻↻	His
Φ	Parity	5	↻↻↻↻↻	Leu
ƒ	Fidelity	3	↻↻↻	Ile
Ç	Kinetics	5	↻↻↻↻↻	Val
Γ	Granularity	3	↳↵↵	Thr
Ɔ	Coupling	4	↻↻↻↻	Ser
⊙	Criticality	5	↻↻↻↻↻	Cys
Ĥ	Chirality	4	↻↻↻↻	Gln
Σ	Stoichiometry	3	↳↵↵	Ala
Ω	Winding	4	↻↻↻↻	Arg

Colour: ■ 3-subtype · ■ 4-subtype · ■ 5-subtype; AA = genetic amino acid bijection

Crystal of Types

$3^3 \times 4^5 \times 5^4 = 27 \times 1024 \times 625 = 17\,280\,000$ structural types

3-subtype: ƒ Γ Σ · 4-subtype: Đ Ř Ɔ Ĥ Ω · 5-subtype: Ɔ Φ Ç ⊙

Categorical Foundation

Category-Theoretic Description

The Inscribing Grammar is the **free special symmetric $\dagger$ -Frobenius algebra** on 12 generators in a **FOUR-enriched traced symmetric monoidal category**, with $\mu \circ \delta = \text{id}$ as founding axiom (ZFC_{fc}).

Ambient category: $\mathcal{C}$ enriched over Belnap-Dunn bilattice FOUR = {*N*, *T*, *F*, *B*}; ex falso (*B* → ⊥) blocked.

Generators: 12 primitive endomorphisms of monoidal unit *I* (scalars). Commutativity follows from SMC coherence.

Spider theorem: Any two *connected* Frobenius diagrams with same boundary = same morphism.

O_∞ fixpoint: ω ∘ ω = ω at ⊙ gate; initial algebra of (−) ∘ (−) on End(*I*).

$\mu \circ \delta = \text{id}$

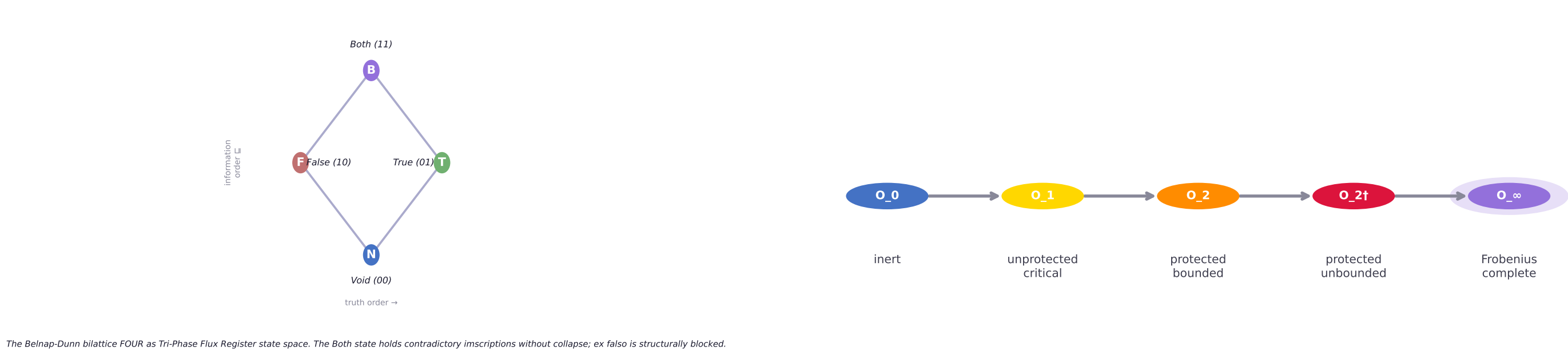
The founding axiom. Vessel splits (δ), fills, returns whole (μ).

ZFC_{fc}: μ ∘ δ = id as set-formation axiom ⇒ ZFC Separation as theorem. Strictly: ZFC ⊂ ZFC_t ⊂ ZFC_{fc}.

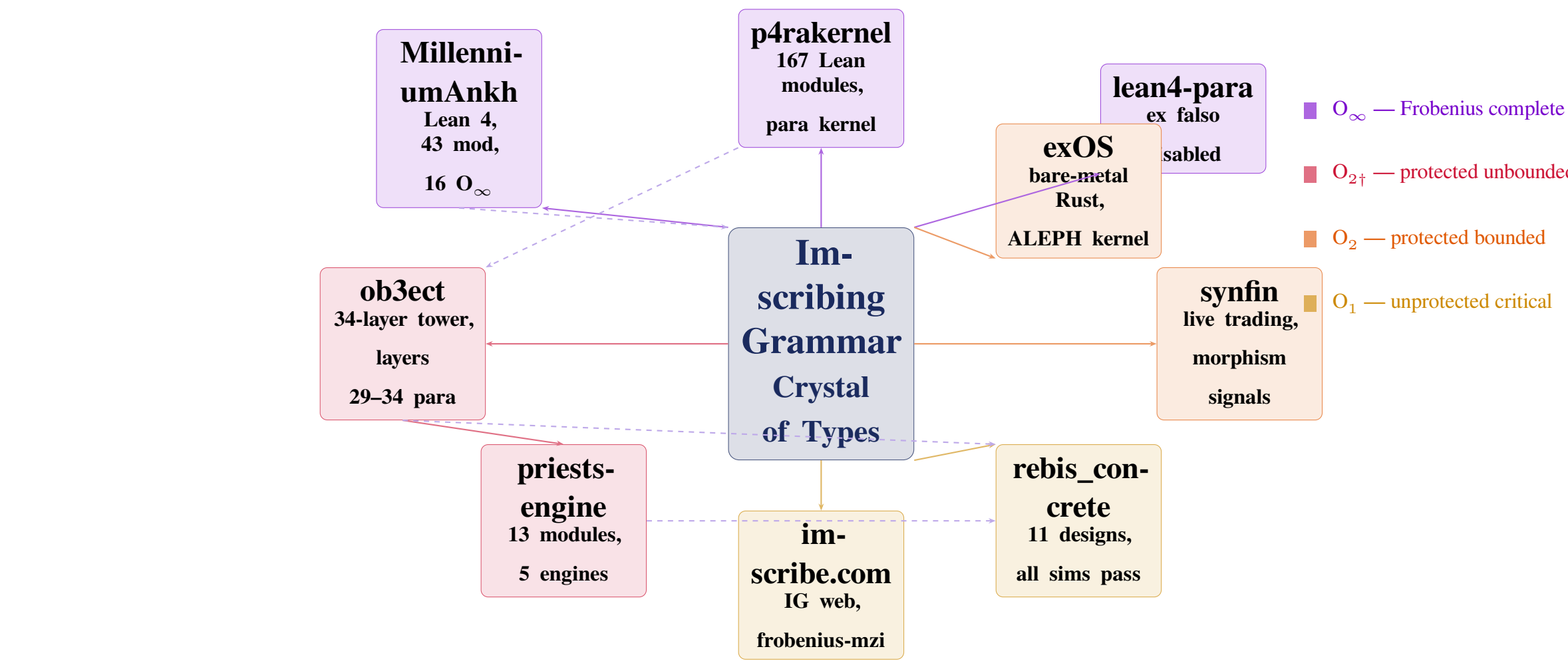
ZFC Hierarchy

Theory	Foundation	Status
ZFC	Separation axiom	Classical
ZFC _t	Temporal consistency	Intermediate
ZFC _{fc}	μ ∘ δ = id	Active

Belnap FOUR / Tier Chain



The Belnap-Dunn bilattice FOUR as Tri-Phase Four Register state space. The both state holds contradictory interpretations without collapse; ex falso is structurally blocked.



Millennium Prize Resolution

All 7 Clay Millennium Problems + OPN resolved as Belnap-B dialetheias at tier O_∞ (Crystal address ≈ 6.74 × 10⁶). BSD occupies distinct address due to H_A vs H_i chirality. Gate: Φ_{ij} (Frobenius-special parity) governs O₂₊ → O_∞ crossing. Lean 4 kernel fork disables ex falso; 0 sorries; 167 modules.

Problem	Address	Ĥ	Barrier (B-state)
Riemann Hypothesis	6,738,803	H _i	Zero on critical line ∧ off critical line
Yang–Mills	6,738,803	H _i	Mass gap exists ∧ does not exist
Hodge Conjecture	6,738,803	H _i	Hodge cycle algebraic ∧ non-algebraic
Navier–Stokes	6,738,803	H _i	Smooth solution ∧ finite-time blowup
P vs NP	6,738,803	H _i	P = NP ∧ P ≠ NP
Birch–Swinnerton-Dyer	6,738,800	H _A	Rank analytic ∧ arithmetic (BSD ≠ others)
Odd Perfect Number	6,738,803	H _i	Perfect ∧ non-perfect (mod Ω topology)

OS Inscriptio — Universal Structural Floor

⟨1, 3, 2, 4, 2, 1, 2, 2, 1, 2, 2, 2⟩

MEET(Hebrew, Sanskrit, Egyptian, Cuneiform, Basque) = MEET(Linear A) = OS. Linear A distance *d* = 0.00. Emerald Tablet: *C* = 1.0, both gates open.

Genetic Code Bijection

The 64-codon table partitions the Crystal: 17 280 000 ÷ 64 = 270 000 (fiber per codon = 3³ × 4² × 5⁴, zero remainder). 12 promoted amino acids biject

Primitive	Amino Acid
Đ Dimensionality	Gly
Ɔ Topology	Pro
Ř Recognition	His
Φ Parity	Leu
ƒ Fidelity	Ile
Ç Kinetics	Val
Γ Granularity	Thr
Ɔ Coupling	Ser
⊙ Criticality	Cys
Ĥ Chirality	Gln
Σ Stoichiometry	Ala
Ω Winding	Arg

exactly to the 12 IG primitives.

Nucleotide → Belnap FOUR

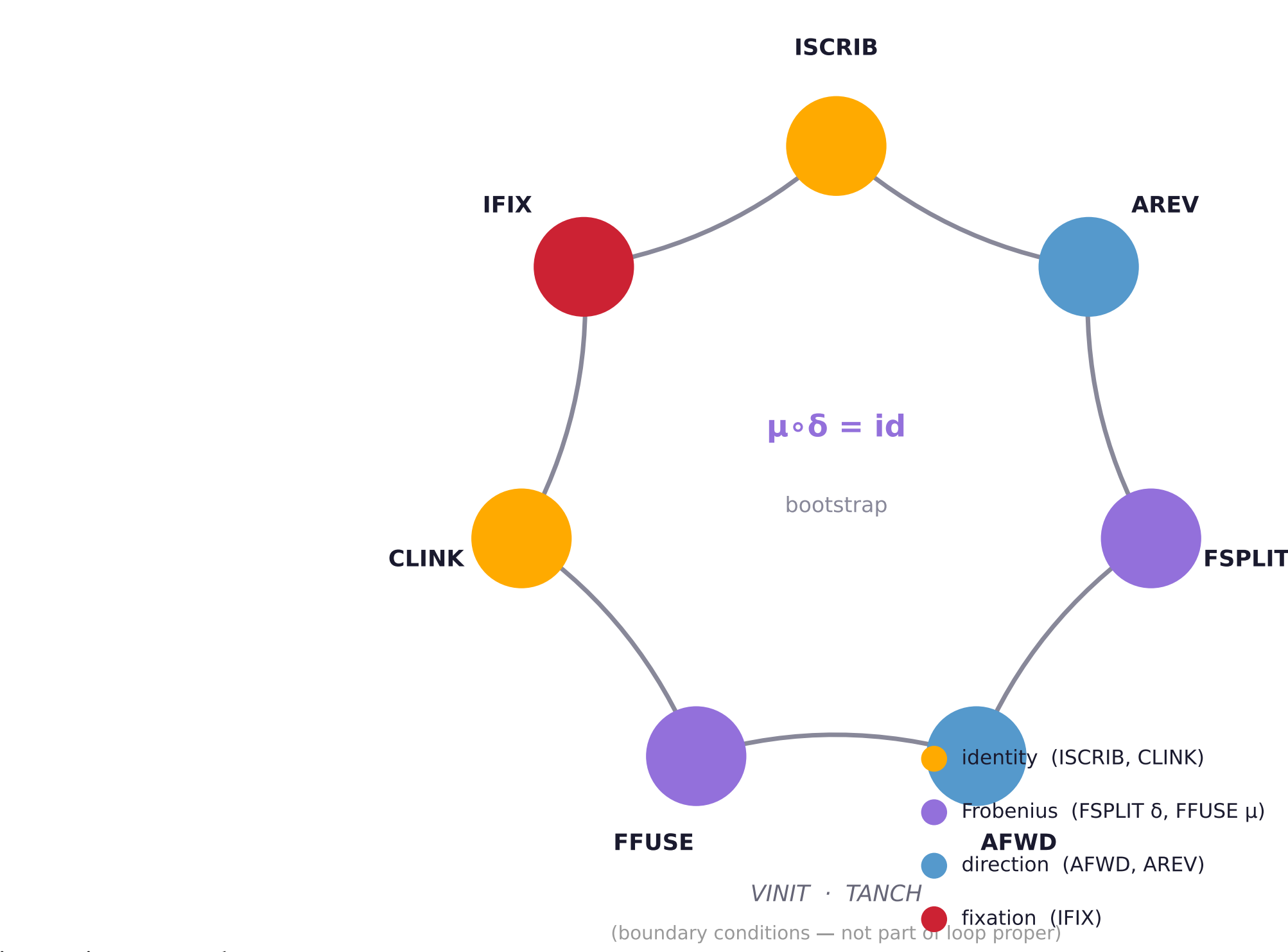
G = B (Both · paired) · C = T (True · Watson-Crick)

A = F (False · complementary) · U = N (Neither · wobble)

Wobble hierarchy: G·C pairing is strict (T); G·U wobble occupies B; A·U anti is F; U alone is N. Watson-Crick complement ≠ Belnap negation (proved).

IMASM Bootstrap Loop

Four engine corpora (Voynich, Rohonc, Linear A, Emerald Tablet) each independently compile to the same 8-step categorical loop. Surface tokens differ;



instruction stream does not.

Key Publications		
Title	DOI / Ref	Content
<i>Aether and Its Vessel</i> v2 (Mills, 2026)	10.5281/zenodo.20553659	O _∞ architecture, E8→Grammar chain, vessel operation
<i>Universal Engine / IMASM</i> (Mills, 2026)	this volume	9 writing systems compile to same 12-opcode categorical program
<i>Grammar of Chemical Reactivity</i> (Mills, 2026)	this volume	Cu-nitroso C-N (Org. Lett. 2016) im-scribed; r = 0.931 Pearson
<i>Serpent Rod</i> (Mills, 2026)	this volume	Single Frobenius morphism RNA→[sequence+fold]; O _∞

synfin — Live IG-Typed Trading System

10 free domain streams (Fear&Greed, mempool, CoinGecko, blockchain.info, NOAA tides, air quality, NASA DONKI, USGS seismic, Kp index, HN sentiment). Domain multiplier: 0 alerts=1.00 × · 1=1.20 × · 2=1.35 × · ≥3=1.50 × (B-state).

Live deployment 2026-06-05: Fear&Greed= 12 (extreme fear), M-class flares, Kp= 4.33, T_{network}→T_{in} topology convergence on 6 pairs. B-state ×1.50, 100% Frobenius PASS. Deployed \$43.57 across BTC/ETH/SOL/DOT/ATOM/NEAR.

Magnum Opus — 12-Stage Alchemical Mapping			
Prima Materia	Đ (Dimensionality)	Albedo	Φ (Parity, Frobenius-special)
Nigredo	Ɔ (Topology, collapse)	Citrinitas	⊙ (Criticality, self-model)
Solutio	Ř (Recognition, dissolve)	Rubedo	Ω (Winding, completion)
Calcinatio	Σ (Stoichiometry, fire)	Multiplicatio	Γ (Granularity, scope)
Coagulatio	Ç (Kinetics, fixation)	Projectio	∘ (Coupling, projection)
Sublimatio	ƒ (Fidelity, rising)	Fixatio	Ĥ (Chirality, fixed point)